a Three sensors, three band lists (native-v1 shards)

EnMAP

bands	224, each scene's own
aligned to	enmap_224.json
pixel	30 m (tile 3.84 km)
train / test	14,542 / 950
split by	datatake (6 in test)
labels	cloud, cirrus 0-3, haze, shadow, snow, land/water
quirk	QL_PIXELMASK gaps

PRISMA

bands	234 of 239 (flags)
aligned to	none needed
pixel	30 m (tile 3.84 km)
train / test	14,802 / 1,570
split by	15 km site
labels	none
quirk	VNIR listed 980 nm down

AVIRIS-NG

bands	372 of 425 (bbl)
aligned to	aviris_ng_372.json
pixel	3.2-8.2 m (0.4-1 km)
train / test	22,377 / 2,270
split by	15 km site
labels	none
quirk	4 product versions

b Alignment by wavelength, 0.5 nm tolerance (band_alignment.py)

Every training batch:

one sensor, drawn by share
EnMAP 0.40 / PRISMA 0.35 / AVIRIS-NG 0.25
every sample aligned to that sensor's list
so the stem gets one (N,) wavelength list